

02

NETWORKING

7 LESSONS

PHASE 1 • FUNDAMENTALS

Networking & Protocols

TCP/IP, DNS, HTTP, routing, and the protocol stack – the language every attack and defense speaks across the wire.

01

The OSI Model & TCP/IP Stack

BEGINNER

CONCEPT

25 min →

The seven conceptual layers of networking and how they map to the rea...

02

TCP, UDP & Ports in Depth

BEGINNER

CONCEPT

30 min →

The three-way handshake, sequence numbers, UDP vs TCP, and the port n...

03

IP Addressing, Subnetting & Routing

BEGINNER

CONCEPT

35 min →

IPv4/IPv6 address math, CIDR, subnet masks, and how routers really mo...

04

DNS in Depth & DNS Attacks

INTERMEDIATE

CONCEPT

35 min →

Hierarchy, resolution process, record types, and the attacks that use...

05

HTTP, HTTPS & TLS Deep Dive

INTERMEDIATE

LAB

40 min →

Request/response model, headers, methods, cookies, and the TLS handsh...

06

Wireshark & Packet Analysis

INTERMEDIATE

LAB

45 min →

Capture, filter, decode and analyze network traffic – the #1 skill fo...

07

Firewalls, IDS & IPS

INTERMEDIATE

LAB

40 min →

The first line of network defense: rules, stateful filtering, intrusi...

← Hardware, Computing & IoT Fun...

Linux, Windows & Operating Syst...